Flooring Subfloor Build Process

Scope

- Define the locked floor stack-up, hardpoint strategy, procurement state, and install workflow for the truck bed floor.
- Baseline date for this revision: May 14, 2026.
- Moisture-control premise updated May 3, 2026: the Hiatus is effectively a sealed capper with barn doors; the inside of the truck bed remains the build substrate, so water must not be allowed under the subfloor.
- Field progress updated May 4, 2026: bed rail caps were re-bonded/sealed with polyurethane; small drill holes and corner gaps were patched/sealed; trowel is purchased.
- Field progress updated May 14, 2026: the subfloor is complete enough to serve as the current working datum, while Lonseal remains intentionally unglued until rough-in, hardpoints, and service-access decisions are validated in the installed camper.
- Owner update 2026-07-06: after all major modules were installed/connected and fitment-verified, flooring moved back up as the likely next gated sprint. Hardpoint determination must happen before module removal and may take at least a full day; Lonseal glue-down waits until those hardpoints are drilled/registered and the plywood/floor stack is ready.

Final Subfloor Structure (Bottom to Top)

1. Existing spray-in bedliner.
2. `5/8 in` SilveRboard Graphite EPS between ribs/valleys only. Current state: installed/trimmed enough for the current subfloor datum; inspect for proud/high spots, rocking, moisture paths, or hardpoint conflicts before finish-floor closure. EPS is not a continuous bearing layer and normally does not need to be removed at rib-supported plywood areas, but do not use foam as the clamp-load bearing member at a structural anchor.
3. `3/4 in` birch plywood over the bed rib/EPS valley structure. Current state: complete in three slices with bottom/edges sealed and top bond face untreated for Lonseal adhesive; keep removable/serviceable until hardpoint drilling/registration and service-access gates pass.
4. Lonseal Lonwood Madera marine sheet vinyl, glue-down finish layer. Current state: sheet vinyl and adhesive are in hand; glue-down is now a gated near-term foundation step after installed-module hardpoints are determined, not generic late cosmetic work.

Explicitly De-Scoped Layer

- Full-bed `EPDM/RPDB` thermal-break sheet above EPS and below plywood is no longer in the active floor stack-up.
- EPDM order was canceled before shipment and refunded; no EPDM material is on hand for this build.

Hardpoint Rule (Structural Anchors)

- Determine hardpoints with the modules still installed/connected and fitment verified. This is now its own pre-teardown work block and may take a full day.
- Default active path: furniture/module floor tabs bolt through the furniture bracket, Lonseal, plywood clearance holes, and any local valley/air/EPS zone into Plusnuts/rivnuts installed directly in the truck bed metal from inside the bed. The Plusnut/rivnut body and flange seat in the steel bed floor, not in the plywood; the plywood/Lonseal are clamped layers above the metal anchor.
- Prefer hardpoint locations on rib/high areas where plywood bears on steel. If a required hardpoint lands over a valley, treat EPS as a pass-through filler only; add/verify a solid clamp-load path rather than relying on foam compression.
- Use the furniture hardpoints as the primary plywood-to-bed clamp where their locations also solve plywood flatness. Add separate plywood-only Plusnuts only where needed to pull down a proud/uneven plywood area, control a seam/edge, or preserve the floor independently of furniture removal.
- Do not rely on screws into plywood for structural furniture retention; screws can loosen/release under vibration. Structural retention should load the truck bed metal via Plusnuts/rivnuts or other approved mechanical anchors.
- Drill/register hardpoint holes before Lonseal. Install/test the Plusnuts, then preserve hole locations through the vinyl by using a measured datum map, photos, marked tape, and/or pre-punched/dry-fit Lonseal holes with a few temporary locator bolts during glue-down.
- Seal/protect drilled plywood holes and exposed metal edges before final assembly.

Procurement Snapshot (Updated)

Purchased / in hand (active flooring path)

- SilveRboard Graphite EPS kit (`$71.38`) - Home Depot.
- Loctite PL 300 foamboard adhesive (`$5.48`) - Home Depot.
- TotalBoat Halcyon Water-Based Marine Varnish, clear gloss, pint (`$22.99`) - Amazon.
- `3/4 in` birch plywood, `4x8` , qty `2` (`$88` each, `$176` total).
- Lonseal Lonwood Madera Topseal marine sheet vinyl.
- Lonseal adhesive.

- Sheet-vinyl adhesive trowel: Roberts `10-181-16` exact-notch trowel (`$4.40` ; BOM row `174`).

Purchased / partially used (bed sealing workstream)

- Loctite polyurethane window/door/siding sealant, `10 oz` , qty `2` at `$9` each (`$18` ; BOM row `177`), used under bed rail caps and at selected corner gaps.
- Gorilla Waterproof Patch & Seal Tape, `8 ft` roll (`$14` ; BOM row `178`), roughly half used on small drill holes and other minor bed openings.

Canceled / refunded (not in hand)

- Rubber-Cal EPDM closed-cell sponge rubber, qty `2` (`$47.22`) - Home Depot order canceled before shipment; refunded.

Related purchase (not floor stack, but same workstream)

- `1/2 in` plywood, `4x8` , qty `1` (`$68`) for electrical backer/closet module.

Not Yet Purchased / confirm before glue-down

- Compact segmented roller or equivalent if not already in hand (`row 175`).
- Fresh utility blades, masking tape, gloves, rags/cleaner, and disposable adhesive cleanup supplies as needed.

Totals

- Active flooring package purchased to date: `$280.25` (adds purchased trowel row `174`).
- Active flooring package remaining expected spend: `$424.68` .
- Active flooring package estimated total: `$704.93` .
- Bed sealing consumables added outside the prior flooring package total: `$32` (`rows 177-178`).

Historical provenance

- INSTALL_MINUS_12_READINESS_PLAN preserves the May 7 install-window floor/seal planning.
- STARTER_PLAN_electrical_and_flooring_pre_camper preserves the pre-camper floor/electrical starter plan.

Installation Sequence

Phase 0 - Bed rail dust/weather closeout before finish floor

Status: historical install-window workstream; inspect after normal travel/shakedown before any finish-floor closure.

1. Empty the bed and remove the existing removable flooring before the Washington trip; do not transport flooring through the wet/adhesive rail-seal workstream. **Completed.**
2. Remove or lift bed rail caps as needed and photograph all old camper-install holes. **Caps removed; most clips broke.**
3. Consider tailgate removal only if it improves access or drying; store it safely and protect wiring/camera connections.
4. Confirm with Hiatus that the old roughly `2 in x 3 in` top-rail holes are not needed for the new camper interface if still relevant to the final attachment method.
5. Clean/degrease rail surfaces and rust-protect bare cut edges.
6. Close holes as dust/weather intrusion points using low-profile, serviceable cover patches where possible: thin aluminum/ABS/HDPE or similar non-absorbent patch/backer bedded in butyl tape, automotive seam sealer, MS polymer, or polyurethane sealant.
7. Reinstall rail caps with targeted sealing/gasketing while avoiding camper-fit interference and avoiding hidden pockets that can hold water against metal. **Completed with large Loctite polyurethane bead under caps.**
8. Let sealant skin/cure per product conditions before reinstalling flooring or handling surfaces. **Historical gate; perform post-travel inspection before closeout.**

Phase 0A - Bed-floor drain/corner sealing before subfloor closeout

Status: partially complete as of May 4; cure/post-drive inspection still required.

1. Treat factory drain holes, unused bed-floor penetrations, and small corner gaps under/near the subfloor as water-ingress defects, not intentional drains for this build.
2. Seal them before final subfloor/Lonseal closeout using reversible or serviceable automotive/marine methods where practical: OEM-style rubber plugs, butyl-bedded aluminum/ABS/HDPE patches, automotive seam sealer, MS polymer, polyurethane sealant, or patch/seal tape for small drill holes where adhesion is clean and protected.
3. Do not rely on household silicone as the primary sealant.
4. Do not seal unrelated exterior body drains or create trapped metal cavities; the target is zero upward water path into the EPS/plywood cavity, while still allowing inspection/service before the finish floor is glued.

5. Field status: small drill holes and several other minor bed openings were sealed with Gorilla Waterproof Patch & Seal Tape; selected corner air gaps received small polyurethane beads.
6. After sealant cure, camper install, and normal travel/shakedown, inspect the bed floor/corners and inside of the bed before reinstalling EPS/plywood. No dedicated hose/leak test is required for this under-subfloor path; the concern is ordinary underside spray/dust, not submersion.

Phase 1 - Prep / current-state correction

1. With modules still installed/connected, spend the hardpoint day before teardown: locate each floor tab/bracket, test access for drill/Plusnut tool/bolt head, and mark candidate hole centers.
2. Use transfer punches/tape/photo references to capture hole centers, module footprints, seams, shims, battery extraction path, and service-panel zones before anything moves.
3. Prefer anchor centers on rib/high areas where the plywood is already supported by steel. If an anchor must land over a valley/EPS zone, solve the clamp-load path rather than depending on foam.
4. Drill pilot/registration holes through the plywood/truck-bed stack only after checking below/behind for hazards and confirming the bolt head will remain accessible after reinstall.
5. Clean bed thoroughly after module removal.
6. Trim only EPS proud/high spots that interfere with plywood seating; EPS in valleys does not need blanket removal.
7. Confirm the three plywood slices sit flat enough for adhesive and final module reinstall.

Phase 2 - Between-Rib Insulation (mostly complete)

1. EPS remains between ribs/valleys only.
2. Maintain light tack-only adhesion where used; avoid full-surface bonding.
3. Confirm EPS does not create proud spots that rock the plywood.
4. At hardpoints, EPS may be pierced by the bolt/hole path, but it is not the structural clamp member. Prefer supported rib locations or add/verify a non-foam clamp path where a valley anchor is unavoidable.

Phase 3 - Subfloor (`3/4 in` birch, three-slice state)

1. Verify panel seam strategy and removal path before finish flooring.
2. Confirm sealed bottom/edges/holes and untreated clean top bond face.
3. Drill/register all chosen module hardpoints before Lonseal while the hole map is visible and recoverable.
4. Install and test Plusnuts/rivnuts in the truck bed floor from inside the bed where under-bed access is impractical.
5. If new anchor holes are drilled, seal the inside of all plywood holes and protect exposed metal edges before final install.
6. Use furniture hardpoints as the primary plywood clamp where they land well; add separate plywood-only Plusnuts only where needed for flatness/seams/edges.
7. Reinstall plywood and verify bolt access, service openings, and battery extraction before glue-down.

Phase 4 - Lonseal glue-down gate

Do not proceed until:

- bed rail, bed-floor drain-hole, and corner-gap sealing is complete and visually inspected after cure/normal use, not merely deferred;
- modules have been used to determine actual hardpoint locations before teardown;
- hardpoint pilot/Plusnut holes are drilled, deburred, protected, test-threaded, and mapped from fixed datums;
- Lonseal hole strategy is chosen: pre-punch/cut holes during dry-fit, use a few temporary locator bolts/pins during layout, or preserve a highly reliable datum map before final cutting;
- EPS trim and plywood seating pass, including any proud/high valley filler corrections;
- no floor-through penetrations, drains, heater routes, anchor holes, or hidden hardpoints need access under the finish sheet unless intentionally registered/reserved;
- top plywood face is clean, dry, untreated, and adhesive-compatible;
- exact trowel is purchased and roller/rolling method is confirmed;
- dry cure window is available.

Phase 5 - Finish Flooring

1. Dry-fit Lonseal sheet and align grain/pattern.
2. Transfer known Plusnut/hardpoint centers to the Lonseal during dry-fit; punch/cut clean fastener holes before final glue where practical.
3. Apply adhesive per Lonseal adhesive manufacturer specification using the exact required trowel.
4. Lay vinyl and roll thoroughly, using only the minimum temporary locator bolts/pins needed to keep critical hardpoint holes aligned.
5. Trim edges and allow cure before loading furniture.
6. Protect the cured sheet during module reinstall, then bolt furniture through the registered Lonseal/plywood stack into the tested Plusnuts.

Final Build Height

- EPS: `0.625 in`
- Plywood: `0.750 in`
- Vinyl: `0.080 in`
- Total: approximately `1.46 in`

Confirm final door and tank clearances against this stack-up before permanent installation.

Engineering Notes

- Thermal performance: moderate floor thermal decoupling from EPS; less top-side thermal break than prior EPDM-inclusive concept.
- Structural behavior: stiffer walking surface and higher point-load robustness with `3/4 in` birch. Furniture should not be structurally screwed only to plywood; the active retention path is mechanical anchors into truck-bed metal, with plywood/Lonseal treated as clamped layers in the stack.
- Moisture durability: sealed bottom face, edges, and drilled holes remain required because wood is directly above rib cavities; the top face stays untreated to remain compatible with the Lonseal vehicle-install substrate requirement for untreated plywood.
- Bed-floor water policy: the subfloor cavity is not a drained cavity in this build. Because the camper has sealed barn doors and the bed floor remains the build substrate, factory drain holes and small corner gaps that can admit water under the subfloor should be sealed and visually inspected before finish-floor glue-down; no dedicated hose/leak test is required for ordinary underside-spray risk.
- Hardpoint serviceability: under-bed drilling/access is expected to be impractical, so hole drilling, Plusnut setting, and hole registration should be planned from inside the truck bed before Lonseal hides the holes.
- Acoustic behavior: no dedicated EPDM damping layer in active stack; expect more direct structure-borne vibration vs prior concept.
- Serviceability: finish vinyl remains the only permanent bond layer, but one-piece Lonseal glued over three plywood slices will make subfloor removal materially harder. Treat glue-down as an explicit gate, not casual finish work.